

1. Administrative & Study Identification

Full Study Title: Phase III Study of Efficacy and Safety of Secukinumab Versus Placebo, in Combination With Glucocorticoid Taper Regimen, in Patients With Polymyalgia Rheumatica (PMR)
Official Title: A randomized, parallel-group, double-blind, placebo-controlled, multicenter Phase III trial to evaluate efficacy and safety of secukinumab administered subcutaneously versus placebo, in combination with a glucocorticoid taper regimen, in patients with polymyalgia rheumatica (PMR)
Short Title/Acronym: REPLENISH **Protocol Number:** CAIN457C22301 **NCT Number:** NCT05767034 **Posting ID:** 3728312649095137 **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Sponsor/Company Name:** Novartis Pharmaceuticals (Novartis Pharma AG)
Investigational Product: Secukinumab (Cosentyx®) **ATC Code:** L04AC10 **Phase of Development:** Phase III **Medical Monitor:** Daniel Monney, Novartis Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To demonstrate the efficacy of secukinumab 300 mg subcutaneously in combination with a 24-week glucocorticoid (GC) taper regimen compared with placebo with respect to the proportion of patients in sustained remission at Week 52. **Secondary Objectives:** To assess the difference in the proportion of patients achieving complete sustained remission at Week 52, the adjusted annual cumulative GC dose, the time to first use of escape or rescue treatment through Week 52, and the evaluation of Patient-Reported Outcomes (PROs) and overall safety.

2.2 Background & Rationale Polymyalgia rheumatica (PMR) is the second most common inflammatory rheumatic disease in adults aged 50 years and older. Current standard-of-care treatments rely heavily on long-term glucocorticoids (steroids), which carry severe risks such as osteoporosis, diabetes, and other steroid-associated toxicities. Because up to 40% of PMR patients experience relapse during steroid tapering, the REPLENISH trial evaluates secukinumab (an IL-17A inhibitor) as a steroid-sparing therapy to maintain sustained remission, reduce long-term glucocorticoid exposure, and improve patient quality of life.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1

(Secukinumab 300 mg : Secukinumab 150 mg : Placebo) **Disease Area:** Polymyalgia Rheumatica

3.2 Planned Enrollment & Duration Target Enrollment: 381 patients
Number of Sites: Multicenter (Conducted across 27 countries globally)
Estimated Study Start: 2023 **Estimated Primary Completion:** Q4 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or non-pregnant, non-lactating female participants of at least 50 years of age. 2. Diagnosis of PMR according to the provisional ACR/EULAR classification criteria (history of bilateral shoulder pain accompanied by elevated CRP ≥ 10 mg/L and/or elevated ESR ≥ 30 mm/hr, and scoring at least 4 optional points). 3. History of being treated for at least 8 consecutive weeks with prednisone ≥ 10 mg/day (or equivalent). 4. At least one episode of PMR relapse (recurrence of bilateral shoulder/hip girdle pain associated with inflammatory stiffness and elevated ESR/CRP) while attempting to taper prednisone at a dose ≥ 5 mg/day within 12 weeks prior to Baseline. 5. Treated as per local treatment recommendations following the latest PMR relapse and currently on prednisone of at least 7.5 mg/day and not exceeding 25 mg/day at screening.

4.2 Exclusion Criteria 1. Evidence/history of Giant Cell Arteritis (GCA) as indicated by typical cranial symptoms, extremity claudication, imaging, and/or temporal artery biopsy result. 2. Relapse or symptoms driven by other diseases that mimic PMR, such as osteoarthritis, polyarticular calcium pyrophosphate deposition disease, rotator cuff disease, adhesive capsulitis, or fibromyalgia. 3. Concurrent rheumatoid arthritis, other inflammatory arthritis, or connective tissue diseases (e.g., SLE, systemic sclerosis, vasculitis, myositis, ankylosing spondylitis). 4. Concurrent diagnosis or history of neuropathic muscular diseases or fibromyalgia. 5. Inadequately treated hypothyroidism. 6. Previous exposure to secukinumab or other biologic drugs directly targeting IL-17 or IL-17 receptor. 7. Treatment with tocilizumab or other IL-6/IL-6-receptor inhibitors within 12 weeks or within 5 half-lives (whichever is longer) prior to Baseline.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Secukinumab 300 mg or Secukinumab 150 mg versus matched Placebo. Administrations will occur once every week for the first 4 weeks, and once every 4 weeks thereafter. **Route of Administration:** Subcutaneous (SC) via pre-filled syringe. **Duration of Treatment:** 52-week treatment period (with the last investigational medicinal product administration at 48 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: A protocol-specified 24-week prednisone (glucocorticoid) taper regimen is permitted and required for all treatment arms. **Prohibited:** Use of other biologic agents, unapproved immunosuppressants, or deviations from the rescue medication guidelines prior to meeting the escape criteria.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to -42 to 0	Informed Consent, Medical History, PMR Symptom Baseline
Baseline	Day 0	Randomization, First SC Injection, Initiate 24-week GC Taper
Interim	Weeks 4, 12, 24, 36	Safety Labs, PMR Assessment, PRO Collection, Taper Compliance
End of Treatment	Week 52	Final Efficacy Assessment (Sustained Remission), PROs
Follow-Up	Weeks 52 to 76	Treatment-free Safety Follow-up up to 24 weeks

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants achieving sustained remission at Week 52. Sustained remission is defined as achieving remission at Week 12 and maintaining it from Week 12 to Week 52 with no recurrence of PMR signs/symptoms requiring escape/rescue treatment, and no new diagnosis of GCA requiring escape/rescue treatment. **Measurement Method:** Clinical assessment of PMR symptoms alongside an absence of protocol-defined escape or rescue therapy use.

7.2 Secondary & Exploratory Endpoints

1. Proportion of patients achieving complete sustained remission at Week 52.
2. Adjusted annual cumulative glucocorticoid (GC) dose adjusted by duration of study follow-up.
3. Time to first use of escape treatment or rescue treatment through Week 52 (measured in days).
4. Steritas Glucocorticoid Toxicity Index (GTI) assessment to evaluate the burden of steroid exposure.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection of key safety data, PROs, and monitoring for injection-site reactions or infections. **Serious Adverse Events (SAEs):** Reported and tracked globally throughout the 52-week treatment and 24-week follow-up phases. **Data Monitoring Committee (DMC):** Utilized by Novartis to independently evaluate ongoing safety data and steroid toxicity profiles.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** A target enrollment of 381 patients is powered to demonstrate statistical superiority of secukinumab 300 mg over placebo in sustained remission rates ($\alpha = 0.05$). **Handling of Missing Data:** Patients discontinuing treatment early or requiring premature escape/rescue medications are generally imputed as non-responders for the primary sustained remission endpoint.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Combination of central, remote, and onsite monitoring across the 27 participating global countries. **Audit Trail:** Strict documentation of the 24-week GC taper compliance and precise tracking of rescue medication deployments.

11. Study Identifiers & Governance

Primary ID: CAIN457C22301 **Registry Number:** NCT05767034 **Posting ID:** 3728312649095137 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** REPLENISH (Phase III Study of Efficacy and Safety of Secukinumab Versus Placebo, in Combination With Glucocorticoid Taper Regimen, in Patients With Polymyalgia Rheumatica)

12. Clinical Framework (PMR Specific)

Primary Condition: Polymyalgia Rheumatica (PMR) experiencing relapse **Diagnosis Threshold:** Recurrence of bilateral shoulder/hip girdle pain with inflammatory stiffness within 12 weeks of Baseline, meeting ACR/EULAR criteria. **Medical Methodology:** Interleukin-17A (IL-17A) inhibitor alongside a 24-week Glucocorticoid taper **Optimization Target:** Sustained clinical remission at 52 weeks and significant reduction in cumulative GC dose

13. Study Procedures & Methodology

Management Pathway: Targeted biologic intervention combined with protocol-dictated steroid tapering **Education Protocol (HCP):** Site training regarding strict adherence to the 24-week GC taper and protocols for declaring treatment escape/rescue. **Education Protocol (Patient):** Self-administration training for secukinumab

subcutaneous injections via pre-filled syringe and daily steroid taper compliance tracking. **Quality Audit Mechanism:** Routine evaluation of the Steritas Glucocorticoid Toxicity Index (GTI) and standardized PRO symptom trackers.

14. Observation & Follow-Up Schedule

Total Duration: Up to 82 weeks (up to 6 weeks screening, 52 weeks treatment, up to 24 weeks safety follow-up) **Onsite Visit Cadence:** Scheduled clinic visits at Baseline, critical interim taper milestones (e.g., Wk 24), and End of Treatment (Wk 52) **Remote Monitoring Frequency:** Intermittent remote tracking corresponding with PRO diary entries and AE reporting **Checklist Review Interval:** Continuous monitoring of the GC tapering sequence and symptom relapse triggers

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	06-Apr-2026				
Clinical Operations Lead	Daniel Monney	_____	06-Apr-2026				
Lead Statistician	[Name]	_____	06-Apr-2026				